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RESEARCH ARTICLE

FFM: Flood Forecasting Model Using Federated Learning

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ABSTRACT Floods are one of the most common natural disasters that occur frequently causing massive damage to property, agriculture, economy and life. Flood prediction offers a huge challenge for researchers struggling to predict floods since long time. In this article, flood forecasting model using federated learning technique has been proposed. Federated Learning is the most advanced technique of machine learning (ML) that guarantees data privacy, ensures data availability, promises data security, and handles network latency trials inherent in prediction of floods by prohibiting data to be transferred over the network for model training. Federated Learning technique urges for onsite training of local data models, and focuses on transmission of these local models on the network instead of sending huge data set towards central server for local model aggregation and training of global data model at the central server. In this article, the proposed model integrates locally trained models of eighteen clients, investigates at which station flooding is about to happen and generates flood alert towards a specific client with five days lead time. A local feed forward neural network (FFNN) model is trained at the client station where the flood has been expected. Flood forecasting module of local FFNN model predicts the expected water level by taking multiple regional parameters as input. The dataset of five different rivers and barrages has been collected from 2015 to 2021 considering four aspects including snow melting, rainfall-runoff, flow routing and hydrodynamics. The proposed flood forecasting model has successfully predicted previous floods happened in the selected zone during 2010 to 2015 with 84 % accuracy.

INDEX TERMS Hydraulic, meteorological, flood forecasting system, federated learning, feed-forward neural network.

I. INTRODUCTION

In recent years, rate of natural and man-made disasters has increased in the world [1]. Global flood risk has raised due to hydrological extremities, increased urbanization and global warming [2]. Floods are devastating natural disasters that result in severe life losses, significant destruction of infrastructure, agriculture and downfall of overall socioeconomic system of a country. Floods are common in all parts of the

world but their intensity vary from region to region [3]. In developing countries, flood occurrences inflict countless casualties every year and cause cruel economic crises, rising pecuniary problems [4]. Global temperature escalation resulting in overall climate change cause an increased rate of snow melting and precipitation due to which floods are becoming more frequent and intense [5]. Figure 1 shows that frequency of flood occurrence in Pakistan is higher than other natural disasters [6]. Floods have been observed to outnumber at all other calamities happened in the South Asian countries during 2021 [7]. To protect life and economic infrastructure,

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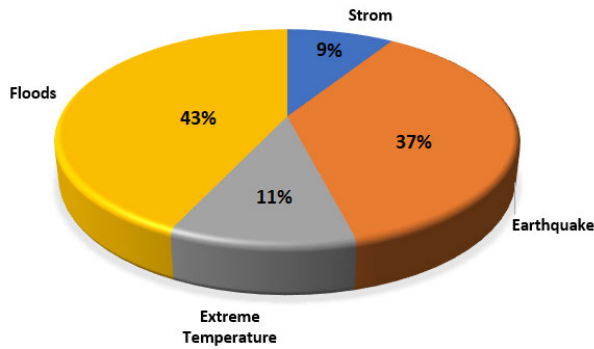


FIGURE 1. Natural hazards percentage as per occurrence in Pakistan during 2021 [6].

governments intensely need a reliable system that may predict floods before its occurrence so that necessary actions might be taken in time [8].

Numerous regionally or globally applicable methodologies, models, and strategies have been put forth but this natural disaster remained hard to model and predict [9]. Despite of best efforts, prediction accuracy has not significantly improved due to varying environmental factors [10].

Many statistical methods and techniques like the climatology average method (CLIM) [11], flood frequency analysis (FFA) [12], Bayesian forecasting models (BFM) [13], artificial neural networks (ANN) [14] and many more are used for flood prediction in literature. In these models, flood causing physical processes have been expressed through complex mathematical expressions.

Machine learning (ML) methods highly contributed in the advancement of prediction systems by providing better performance and cost-effective solutions. Due to the vast benefits and potential of ML, its popularity dramatically increased among hydrologists [15]. Researchers aim to discover more accurate and efficient prediction models through novel ML methods and hybridizing of the existing ones [16]. However, ML based solutions needed huge amount of data to travel on the network for model training. Data is the most valuable asset for every organization [17] and authorities hesitate in sharing data due to various concerns regarding data privacy and data security. Moreover, data protection legislations also prohibit organizations to share data introducing additional barriers in availability of data [18]. Traditionally, intelligent ML techniques have been implemented in flood forecasting that collect and process massive datasets for model training on the server and then disseminate the model towards all clients for making predictions by mutual collaboration. In these techniques, the flood prediction model and data are in a single central location, making the training process centralized and convenient [19]. But centralized setup can hurt the user experience due to back-and-forth communication causing network latency, connectivity delays, battery life losses, data theft and other unpredictable security and privacy issues [20].

It is the need of the time to present an unconventional flood prediction system that can not only predict flood but also

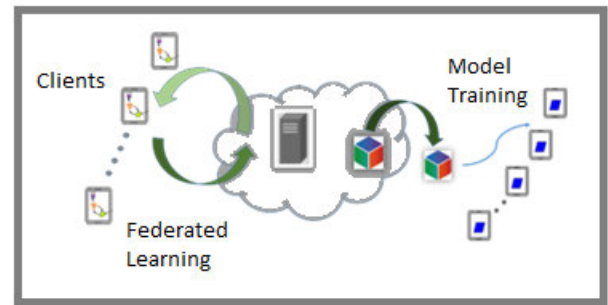


FIGURE 2. Federated learning application.

enforce policies and legislations for preserving data privacy, guaranteeing data security, ensuring data availability, controlling data abundance, and handling latency in data communication [21], [22]. Federated Learning (FL) is the most recent technique of ML that do not need data to be transferred from its origination place rather empowering its clients to perform some pre-processing of data onsite and train their own local data model independently using local data. Local data models are transferred to central server through network for aggregation and training of global model. FL based setup reduced communication overhead to a minimum, while at the same time clients have enough data to render a good model [23]. Privacy, security and communication threats offered by traditional ML models have been automatically eliminated as data is kept safe with the owner organization. Moreover, network issues have also been suppressed as the huge dataset is never being transferred over the network rather a small sized model trained from local data is travelling over the network reducing overheads [24]. Privacy and data security issues related to traditional ML based solutions have enforced the application of FL technique for the training of secured and privacy protected flood prediction models. FL facilitated in the training of a decentralized learning model where user data is never sent to a central server [25].

In this article, Flood Forecasting Model (FFM) based on Federated Learning (FL) technique has been proposed. In FFM, multiple clients work together to train local ML based models in collaboration with the central aggregator, allowing training data to be decentralized while ensuring local data privacy and enforcing data security constraints. Proposed FFM works in three steps as shown in Figure 2. The first step is onsite training and transmission of local data models using regional datasets towards central server for model aggregation. The next step, global model is trained based on local modes that calculates multiple parameters and predicts the client station where flood is about to happen with 5 days lead time. In the last step, local feed forward neural network (FFNN) model is trained on that specific client station to calculate expected water level and inform authorities for taking necessary actions regarding flood preparedness, mitigation and recovery.

The main motivation of this research is to save life and control widespread damage towards humanity and infrastructure

due to flood by presenting a computational model that might predict flood before its occurrence. The novelty of this work is that we propose for the first time a combined model based on FL and FFNN for flood forecasting.

This article is composed of six sections. Section I presents the introduction to the flood prediction using federated learning. Section II describes the related work regarding early flood prediction techniques. Section III presents the materials and methods of the paper used for flood prediction. Section IV illustrates experimental verification of the proposed flood forecasting model. Section V discusses the results obtained by the application of proposed methodology. Section VI concludes the paper and provides future direction.

II. RELATED WORK

Flood prediction is complex and challenging due to its dynamic nature and reliance on climate conditions [26]. Changes in weather directly relate to flooding probability due to global warming and melting snow [27]. In literature, various techniques, models, frameworks and methods have been proposed to predict or somehow control vast disasters caused by floods.

Data-driven statistical methods used for flood prediction include autoregressive integrated moving average (ARIMA) [28], multiple linear regression (MLR) [29] Flood frequency analysis (FFA) [30] and regional flood frequency analysis (RFFA) [12] have been proposed. Later, empirical orthogonal function (EOF) [31], Bayesian forecasting models (BFM) [32], quantile regression techniques (QRT) [33], and climatology average method (CLIM) [34] have also been focused for flood prediction. Advanced flood forecasting systems for long-term and short-term prediction of floods are also significant for the generation of early flood warnings. The Hydraulic models of flow have been practiced forecasting rainfall, storms and tsunamis [35]. These models are also used to predict impact of climatic change [36], ocean waves [37] and floods [38]. It was observed that long term prediction models were more successful than short term prediction models in hazard prediction [39], [40]. Water level of rivers and dams has been predicted by extracting multiple features of rainfall recorded using sensors [41]. Numerical weather prediction (NWP) model to forecast precipitation and a hydrological model to simulate the change in water flow with time in the natural hydrologic cycle for the prediction of hydrological response is also significant in flood forecasting research [42]. However, highly transient flooding caused by intense rainfall has not been predicted that enforce the need for 2D hydrodynamic model [43].

With the passage of time and due to the advancement in technology, a need of advanced data-driven ML based models was felt. Time series-based ML models to predict water inflow in dams have also been proposed in the literature [44], [45]. These models were capable to numerically formulate non-linearity of flood due to their data-driven nature without requiring the knowledge of basic physical processes.

It has been observed that data-driven models usually make faster predictions with minimal input using ML [46]. Multiple ML algorithms including decision tree (DT) [47], Random Forest (RF) [48] and Gradient Boost (GB) [49] have been applied to improve fault-tolerant accuracy in flood prediction handling more intricate information by applying their complex algorithms in a short time span [50]. Innovative deep learning-based model has been suggested in [51] that provides accurate and timely flood prediction as compared to traditional models. A Digital Elevation Model (DEM) has been proposed in [52] to determine the extent of flooding in different areas by observing imagery data for speeding up the relief operation.

Although many techniques have been proposed for flood prediction but the issues associated with data privacy, data security and data availability could not be resolved. There is a strong need to develop a model that can not only predict floods but also preserve data privacy, guarantee data security and ensure data availability. FL is the most recent development of ML that can render models preserving data privacy and ensure data security by onsite local model training. In FL based solutions, data is not transmitted over the network rather small size local models trained from onsite data are gathered on central server for training of global data model [53]. We presented a unique FL based model combined with FFNN technique to forecast floods in which local models are trained at data origination place preserving privacy and ensuring data availability. These local models rather than data are transmitted to central server for local model aggregation and training of global model, guaranteeing data security. Global model applied multiple rules to predict flood at particular station where FFNN model is trained to predict water level during expected flood. Proposed model is cost-effective, have no privacy issues and does not violate data protection legislations. We proposed for the first time a combined model based on FL and FFNN for flood forecasting.

III. MATERIAL AND METHODS

Floods are at the top of the list of highly destructive natural disasters that are very complicated to forecast. Flooding is a complex phenomenon due to natural non-linearity and dynamicity, making it quite difficult to predict [54].

In this article, Flood Forecasting Model (FFM) has been developed to forecast floods in multiple rivers and barrages of the selected region. The proposed model is composed of five layers as presented in Figure 3. Flood prediction involve a huge amount of multidimensional data. In physical layer, data collection centers are present at the edge, where sensors have been used to collect and transmit data to the local client station for local model training. These sensors include rain gauge sensors, water flow calculating sensors and water level sensors. In the proposed model hydrological and meteorological dataset of a region selected from Central Asia have been processed. Statistics of data collected for this research has been presented in Table 1.

All data parameters recorded on physical layer are rigorously regulated during preprocessing prior to local model training. Pre-processing of dataset is performed on local stations for dataset normalization, removing corrupt records and handling missing values.

Network layer in Figure 3 is responsible for secured transfer of local data models towards central server. TCP/IP protocol has been used in this layer for ensuring privacy preservation of local models. Network layer establishes asynchronous communication by receiving local data models from clients and transporting them to the server for training global model to predict floods.

The processing layer in Figure 3 is responsible for completing the FL cycle by training local models at client stations, the transmission of local models to the central server, local model aggregation, training of global data model at central server and triggering of expected flood alerts. In the proposed model eighteen clients have been established. Central server collects model updates from eighteen clients to train a global model. To achieve a true sense of FL, eighteen local data models instead of data are transmitted to the central server where model ensemble technique and federated averaging algorithm (FedFlood) are applied to train global data model for triggering alerts about the expected flood in the selected region. Application of FL in the whole process involved intrinsic parameters, including batch size of the local model (b) used in every iteration during the learning process, rate of learning local dataset (η), training rounds needed (e), threshold criteria and time (t) needed for the completion of each training round.

Two-dimensional FL is occurring in our proposed model. First dimension involves on device training of local data models and the second dimension covers training of global data model based on the local data models received from local data stations. During FL process at $t = 0$, ω_0 model is received by client devices indicating the start of local model training. Clients also receive initialized intrinsic FL parameters from the central server accompanying the initial model used for starting of model training process locally. Clients update that initial model received from the central server according to their locally calculated weight matrix ω_1 at local stations. Clients transmit back an updated version of initial data model as $\omega_1 = \text{updated model (p, q, b, e, } \eta)$ with input variable 'p' and output variable 'q'. Second dimension of FL is related to training of global data model at the central server. Server collects updated models from all connected clients in a database and calculates weighted average by applying FedFlood algorithm that we have developed for finding federated averaging as presented below:

Total number of connected clients = N

Total number of participating devices = Z

Data points that have participated = n

Updated weighted model will be calculated as:

$$\omega_{1n} = \frac{(n \times Z)}{N} \quad (1)$$

Time taken in each round can be calculated as

Global model training time = Local model training time + λt

Where ' λt ' presents communication delay at time t .

Now, the FedFlood algorithm is applied to calculate the federated average of weighted updates received at the central server. As discussed earlier, communication is started from central server by sending ω_0 towards clients. All active and connected clients will receive this initial model, calculate their weighted models and update the received model according to their own local calculations. S_t is the set of connected clients that have received initial model from the server at time interval ' t '. One or more than one training rounds are performed by every client to calculate weighted matrix having learning rate η locally. Updated model ω_k is transmitted to central server. ' k ' presents the set of parameters used during training by S_t clients on completion of local model training. Finally, the server applied the FedFlood algorithm to calculate the weighted average matrix from received local updates and trains global data model accordingly for triggering flood alerts.

Algorithm 1 FedFlood

```

1: ServerUpdation module
2:  $\omega_0 = 0$ 
3: for  $R = 1, 2, \dots$  do //  $R$  training rounds
4:  $m = \max(Z, k)$  //  $Z$ : active clients,  $k$ : parameters trained
5:  $S_t = m$ 
6: for  $g \in S_t$  do //  $g$  is an active participant
7:  $\omega_{(t+1)}^k \leftarrow \text{ClientUpdation}(k, \omega_t)$ 
8: end_for
9: // privacy determination
10:  $\omega_{(t+1)}^k = \sum_{z=1}^k \frac{n_k}{N} (\omega_{t+1})^k$ 
11: end_for
12: end_procedure
13: module ClientUpdation ( $k, \omega$ )
14: for  $i = 1$  to  $\varepsilon$  ... do // handling local epoches
17:   for batch  $b \in \beta$  do
18:      $\omega_i = \omega_k - \Delta(\omega)$ 
19:   end_for
20: end_for
21: return  $\omega_k$  // client updates are returned to the server
22: end_module

```

The result of the processing layer is transmitted to the application layer for timely intimation to flood prediction centers and flood information centers to generate early warnings about the flood and take precautions in time to save humanity and infrastructure as much as possible. These agencies use 4G/5G technology to issue flood alerts through connected smart devices.

IV. EXPERIMENTATION

In this research, an experimental setup containing eighteen local clients and one cloud based central server has been

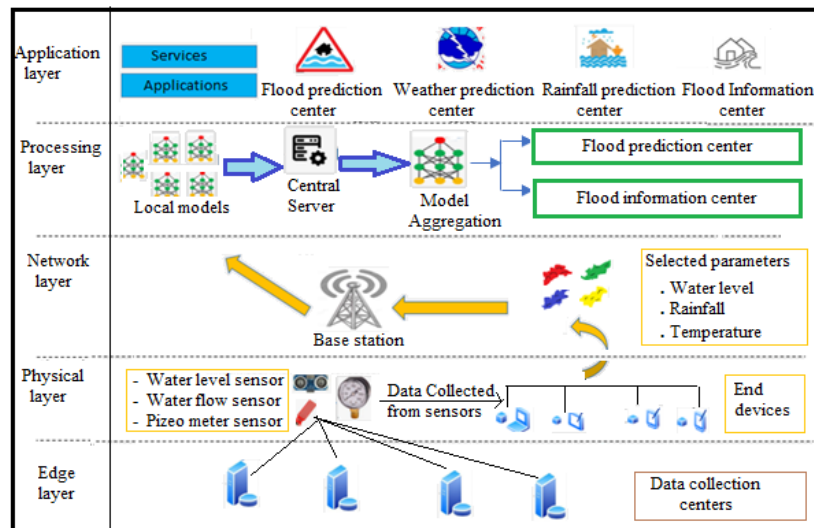


FIGURE 3. Internal structure of the proposed FFM.

TABLE 1. Dataset parameters.

River	Client Stations	Local data Model	# observations	Total capacity (1000 Cusecs) x	Flow Observed (Avg. per month in Thousands of Cusecs)		Danger Level (Thousands of Cusecs)
					In Flow	Out Flow	
River Indus	Tarbela	x_1	100,000	1500	106.0	157.6	650
	Kalabagh	x_2	100,000	950	106.5	153.5	650
	Chashma	x_3	50,000	1000	161.0	150.0	650
	Taunsa	x_4	50,000	1000	158.1	138.4	650
River Jhelum	Mangla	x_5	50,000	1060	17.0	33.9	225
	Rasul	x_6	80,000	850	30.7	11.9	225
River Chenab	Marala	x_7	35,000	1100	36.3	5.0	400
	Khanki	x_8	50,000	1100	7.4	3.6	400
	Qadeerabad	x_9	35,000	900	15.0	9.2	400
	Tarimmu	x_{10}	35,000	875	25.2	9.1	450
	Punjnad	x_{11}	50,000	865	12.0	6.7	450
River Ravi	Jassar	x_{12}	50,000	275	2.9	2.9	150
	Shadara	x_{13}	35,000	250	18.9	18.9	135
	Baloki	x_{14}	35,000	850	32.5	15.8	135
	Sidhnai	x_{15}	50,000	150	16.8	4.7	90
River Sutlej	G.S.Wala	x_{16}	50,000	290	5.8	3.6	50
	Sulemanki	x_{17}	37,000	325	14.6	6.4	175
	Islam	x_{18}	50,000	300	4.3	2.8	175

established. Client data collection centres have been formed on the locations presented in Table 1. FFNN based Flood forecasting module is also installed on every client. The proposed FFM has been implemented in TensorFlow Federated platform using Python. In federated flood forecasting model, central cloud server generates a basic model that is transmitted to all the clients. Each client station is responsible for collecting regional hydrological and meteorological data, making necessary calculations and updating the model received from the central server. Updated models from all clients are transmitted to the cloud server, where federated averaging algorithm (FedFlood) is applied to calculate

the average of all the models received for the generation of updated model. This updated model is then transmitted again to all the clients so that FFM could determine at which station flooding is about to happen in five days lead time.

In the experimental setup, the client is responsible for collecting hydraulic and meteorological data onsite. Edge devices perform pre-processing of data, including data normalization to handle missing, corrupt, and incomplete records. Clients also import necessary libraries of TensorFlow Federated (TFF) for the generation or updating of local data models.

TABLE 2. Dataset bifurcation.

	Observations		
	Total Records	Training set (2015-2019)	Testing set (2020-2021)
Hydraulic dataset	30,000	8000	2000
Meteorological dataset	27,000	8000	2000
Total	57,000	16000	4000

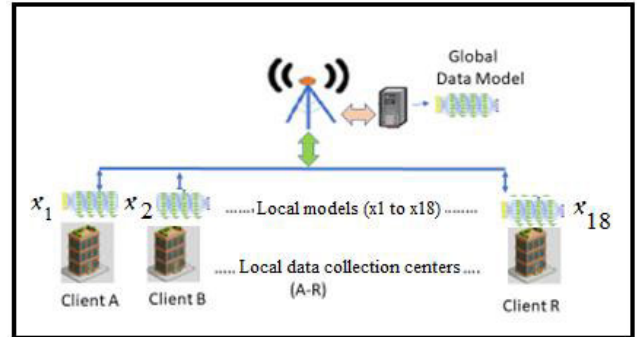
Specification of the dataset used in this experiment has been presented in Table 2. Due to insufficient resources, FFNN model was trained on 20,000 records, including 10,000 records from the hydraulic dataset and 10,000 records from the meteorological data set during experimentation. 20,000 records selected for experimentation were divided into 80 % training set and 20 % testing set given in Table 2.

Client is also responsible to define local data model architecture including batch size(b), learning rate (η) and number of training rounds (e). In the experimental setup, eighteen local data models are trained or updated for transmission of their weight to the central server. The central server applies a federated averaging algorithm (FedFlood) to get the average of local updates sent from clients and uses the results to update the state of global data model through TFF script. FL variables pass through multiple states in the process of training global data model are presented in Table 3. To perform FL for single central server and eighteen clients training script will be executed nineteen times, once at each node. Dictionary parameters from the central server and client keys have been applied to initialize clusters defined in the dataset using `tf.train.CluserSpec()`.

For every iteration, execution of training script starts on central server and will be in “listen and wait” state to finish read/write operation during the specified time lap. The script gets blocked when the waiting time is over, and initialization of server-side FL parameters along with the training process on client hosts would begin. All the clients from x_1 to x_{18} will collectively train a batch onsite and update the weight matrix of every client to ensure an isolated training process. According to the configuration of all clients, weighted model updates are transmitted to the central server where a federated averaging algorithm (FedFlood) is applied to calculate weighted average of all updates received from clients. The newly trained model is shared with all the clients by central server before further batch training. TFF works as a moderator and carefully ensures the synchronization among multiple clients using “replica device setter object”. Central server repeatedly performs the model updating process and instructs the clients to pull new version of the model. This process continues until the threshold defined by model engineer has been achieved indicating that all identified parameters attaining maximum values and the risk of flood is high. After successfully training the global data model, it is transmitted to each client to analyse hydraulic and meteorological data

TABLE 3. State of FL variables in global model training process.

Label	Title of flag
Central Server	Name of Job
Index of task	0-17
Hosting Server	0
Hosting client	Client

**FIGURE 4.** FFM model training process.

through the same model for the prediction of expected flood in the short term. Model training process is presented in Figure 4.

A. COMPONENTS OF PROPOSED FFM

Two major components of FFM have been presented in Figure 5. First component is related to data collection, local model training, sending and receiving models to or from central server. After receiving the generic global model from the central server, flood forecasting module analyses multiple aspects of heterogeneous dataset recorded locally and generate alerts to warn general public or government bodies in advance about an impending flood with as much reliability as possible with 5 days lead time. Flood forecasting module analyses observations of regional multidimensional data about regional rainfall, water elevation or discharge in a specific period ranging from 24 hours to 5 days by training and analysing four different models regarding snow melting, rainfall-runoff, flow routing and hydrodynamics at local stations based upon the received global model. In second component of FFM, results of these four models are collected in a single regional database where rule-based feed-forward neural network technique is applied to calculate expected water level (λ) with five days lead time for making flood forecasts and generating early warnings about the expected flood to save humanity and infrastructure. The Feed forward neural network (FFNN) technique with logistic activation function has been used for forecasting upcoming floods in short term. The Feed-forward technique has been used due to unidirectional nature of signal propagation. It does not involve any feedback cycle between nodes.

B. STRUCTURE OF PROPOSED FFNN MODEL

Proposed FFNN model contained three layers of hidden nodes for transformation of input into output to make flood

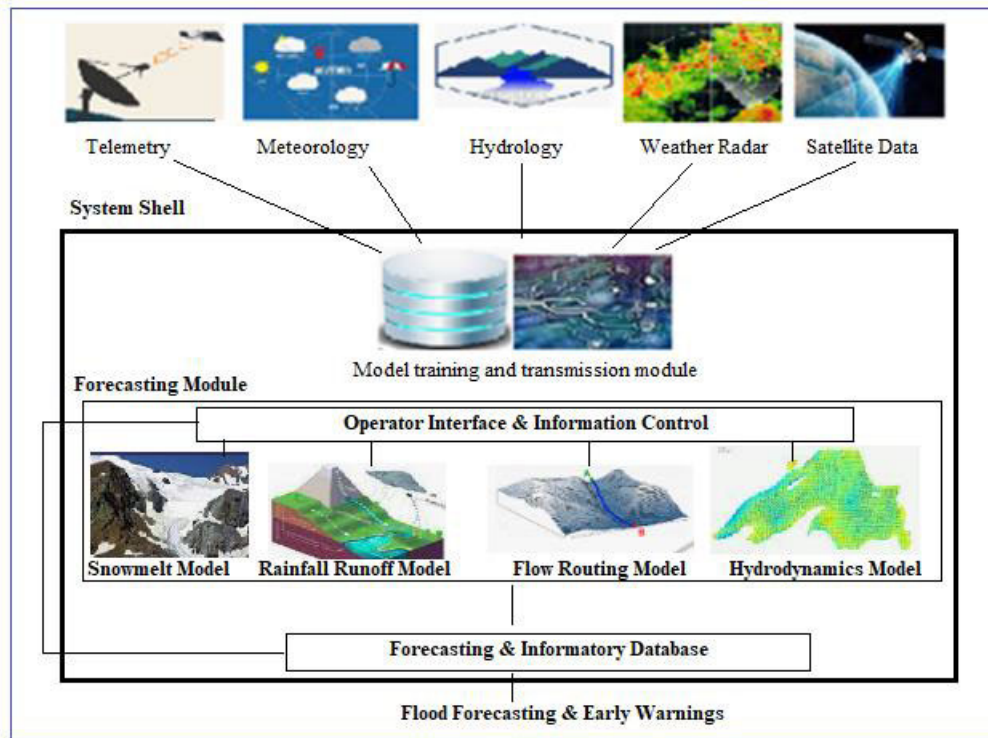


FIGURE 5. Components of the proposed flood forecasting system.

forecast five days ahead. Result of snow melting model, rainfall runoff model, flow routing model and hydrodynamic model becomes input information to be fed into number of inner hidden nodes of FFNN model that works in three steps. In this experiment, initially hidden layer was oversized with 20 nodes. The hidden nodes in first step generates input array using input variable x_i and current water level λ .

$$\text{InArray} = \{x_i, \lambda\} \quad (2)$$

to compute signal z_j from all input information using the following equation.

$$z_j = \sum_i w_{i,j}(x_i - \sigma_j) \quad (3)$$

where z_j is the signal computed by summation of weight $w_{i,j}$ associated with node i, j ; value of the input variable x_i ; and

the threshold value σ . In the second step, hidden layers transform the input z_j using a nonlinear logistic or sigmoid activation function f calculated by the following equation.

$$f(z_j) = \frac{1}{1 + \exp[-z_j]}, f(z_j) \in [0 : 1] \quad (4)$$

In the final stage, the input variable array InArray is transformed into OutArray. Outgoing signal is fed to the output layer of nodes and transformed in the same way as described by the following equations. The transformation is expressed as follows:

$$\text{OutArray} = F(I, \omega, \sigma, \lambda, f) \quad (5)$$

FFNN model training is performed by minimizing the error function $E(\omega)$, representing the square of the error of the model when processing real data. The error of the model is computed during batch training. Training error is calculated as following:

$$\Delta\omega_{i,j} = -\eta \frac{\sigma E}{\sigma\omega_{i,j}} \quad (6)$$

where η is the learning rate. In this experiment, a decreasing learning rate (from 0.5 to 0.1) was used to fast-track convergence to global minimum. Training procedure was repeated multiple times to get best performance. Training was stopped when no further improvement in performance is recorded.

C. MODEL TRAINING PROCESS

For experimentation, structure of FFNN model is defined by trail-and-error method, FFNN model was trained with the intent to reach at minimum error rate $E(\omega)$ for each batch. As short-term flood prediction is focused in this experiment, so $E(\omega)$ on single output node was calculated. The process of minimizing $E(\omega)$ results in minimizing the water level prediction error. Minimum value of $E(\omega)$ is considered as threshold for repetition of training process. Number of hidden nodes was gradually decreased when minimum $E(\omega)$ is achieved, and the training process is repeated. In every iteration, batch of 4,000 records are processed to compute model updates based on the data available locally, then all the updates are averaged at central server. Experiments with data samples containing 4,000, 8000, 12000 and 16,000 records in each

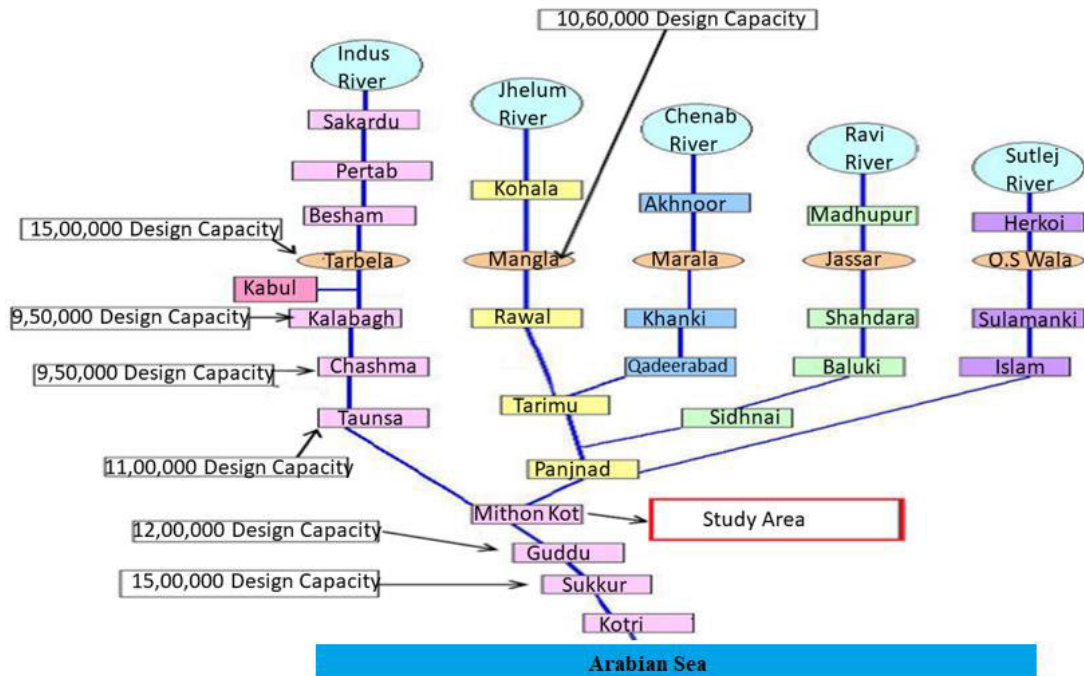


FIGURE 6. Flood route towards the study area located in Punjab, Pakistan [6].

round helped not only in achieving good training metrics for achieving reduced $E(\omega)$ but also handling overfitting in model training as each training round incorporated a different sample with more data to train the model than the previous round. Training process continues until model performance starts decreasing. k-fold cross-validation with $k=3$ (presenting three different data segregations) has also been applied to control overfitting during model training process.

D. PRECAUTION IN FFM DESIGN

While preprocessing, FFS carefully handled data errors, missing data, duplicate instances, unbalanced data through normalization. In the beginning, data have been splitted in training set and test set. Special measures have been taken to avoid sampling bias. During training, FFS defined sampling frame describing the list of stations from where data was processed by fair selection of dependent variables. During validation process, 3-fold cross validation has been opted and in every fold training data have been prepared separately.

V. RESULT AND DISCUSSION

In this experiment, flood possibility at Mithan Kot located in Punjab, Pakistan has been predicted. This area has natural importance in the region because all the major five rivers along with their contributors are collectively passing through Mithan Kot, as presented in Figure 6.

A. INVESTIGATION OF MAJOR FLOOD-CAUSING FACTORS

For experimentation, multiple phenomena have been identified as the main factors causing flooding in five major rivers of Pakistan. The design capacities of dams and barrages built

on these rivers has also been noted to indicate maximum water load tolerance. When incoming water exceeds from their design capacity, then it is evacuated from the dam or barrage causing either an increase in the water level of next dam / barrage or causing flood in lower areas.

Another major cause of flooding identified in this experiment is the high precipitation rate [30]. Heavy precipitation rate is recorded in the regions where rate of rainfall remains high for a long period of time. If R_{avg} is the average rainfall recorded in the region and T presents the time duration of the rainfall, then the total precipitation P_{total} would be

$$P_{total} = R_{avg} \text{ times } T \quad (7)$$

Flooding due to rainfall is associated with high precipitation rate for longer time (in months) whereas instant flooding occurs when rate of rainfall gallops from high to extremely high as presented in Figure 7. Precipitation occurs when moist air is lifted to condensation. Major point of consideration is to determine how much heavy precipitation rate might happen. Therefore, vertical moisture flux (VMF) is calculated that is the ultimate source for precipitation.

VMF is used to calculate precipitation efficiency (γ) as not all the vapours rising high are consumed in cloud formation as presented in Table 4.

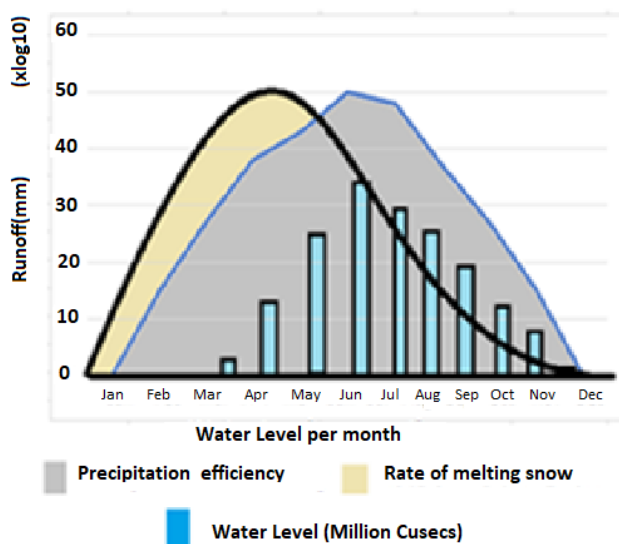
It can be observed from Figure 7 that river water level is directly affected by water added due to precipitation and snow melting. May, June and July are more critical months that add much additional water to the rivers causing floods by the end of July or in the start of August.

FFNN technique has been used for flood prediction in short term (five days in advance) as presented in Figure 8. For short

TABLE 4. Precipitation rate in selected area during 2021 (per month) [2].

Month	Air Velocity (mph)	*Precipitation efficiency $\gamma = \frac{m_p}{m_i}$	Ascent rate (α)	Air mixing rate (β)	VMF $= (\alpha \times \beta)$ (hPa)	precipitation rate $R = \gamma \times \text{VMF}$ (mm)
Jan	14	0.01	16.34	49.27	805.07	23
Feb	9	0.19	14.92	44.14	658.56	28.6
Mar	10	0.25	15.38	45.93	706.40	41.2
Apr	10	0.15	15.43	43.13	665.49	19.7
May	12	0.21	23.45	34.27	803.63	22.4
Jun	19	0.21	29.82	33.5	998.97	36.3
Jul	21	0.61	30.12	23.5	707.82	202.1
Aug	15	0.54	19.21	48.15	925.44	163.9
Sep	18	0.32	16.18	47.56	769.52	61.1
Oct	10	0.13	23.08	36.86	850.72	12.4
Nov	22	0.09	40.75	18.25	743.86	4.2
Dec	24	0.16	13.01	61.52	800.51	13.9

* m_p mass of water in rainfall, m_i mass of water vapors influx in cloud

**FIGURE 7.** Relationship of the average cumulative increase in Indus River water level due to precipitation efficiency and melting snow per month during 2021.

term prediction of flood at specific time 'T', water level at $T-d_1$, $T-d_2$, $T-d_3$, $T-d_4$, $T-d_5$ and amount of rainfall at $T-R_i$ has been calculated. where R_i presents the average rainfall. The temporal trend of rainfall has been used as input for better illustration of basin response. 'T' includes mean lag time of every rain gauge incremented up to current time as rainfall information of previous weeks is also requires for the prediction of expected water level of rivers where clients are working. The transformation of input into output (z_i) defines consecutive patterns of rising limb and falling limb of the flood in temporal window.

B. MODEL EVALUATION

Table 5 summarizes the result obtained by validating the proposed model in terms of accuracy, efficiency and error rate. Proposed model has been tested to predict floods occurred in the selected region during 2010 to 2015 using validation dataset for flood prediction. It can be observed that the values of RMSE are between 0.2 to 0.5 indicates that the proposed model can predict floods accurately [29]. Accuracy of the proposed model has also been calculated from RMSE and it has been observed that the proposed model has predicted the historical floods with 88% accuracy. Model efficiency R^2 is about 99% indicating performance marked up as high satisfactory level [53]. Dispersion diagram of RMSE is shown in Figure 8, where scattering is decreased indicating good performance of FFNN model.

FFNN model's behaviour in time has been compared with six events of Table 5 in Figure 9. It has been noted that true responses (λ_0) against the predicted responses (λ_1) are very close supporting the value of RMSE obtained in Table 5 in the range 0.2 to 0.5 indicating that model can relatively predict the expected flood with reasonable accuracy.

FFNN model predicts floods with 5 days lead time. For making flood prediction water level per day is noted. The error between actual water level and the predicted water level is calculated on every time lap.

Peak height of the water level is also significant in determining model behaviour.

$$\text{Peak height} = |\text{actual height of water level} - \text{calculated height of water level}|$$

Absolute difference of actual and predicted water height is significant in making flood forecasts. Figure 10 presents the

TABLE 5. Model evaluation.

Event	Total Flooded Area (N) (Sq-km)	Year	Water level λ (Cusecs)		Accuracy $A=1.96*RMSE$	Efficiency $R^2=A/(PT * 100)$	Root Mean Square Error $RMSE=\sqrt{(\lambda_0 - \lambda_1)/N}$	Prediction time PT (millisec)
			At Punjnad					
			Actual value (λ_0)	Predicted Value (λ_1)				
A	160,000	2010	310,117	300,000	0.49285816	0.83677108	0.251458247	58900
B	27,581	2011	151,300	145,649	0.88718404	0.898105	0.452644919	98784
C	4,746	2012	65,600	64,707	0.8501938	0.99373947	0.433772347	85555
D	4,483	2013	317,261	316,725	0.67772592	0.812066	0.345778531	83457
E	9,779	2014	454,000	452,874	0.66508636	0.7756471	0.339329777	85746
F	2,877	2015	135,866	135,300	0.86934939	0.9195864	0.443545609	94537

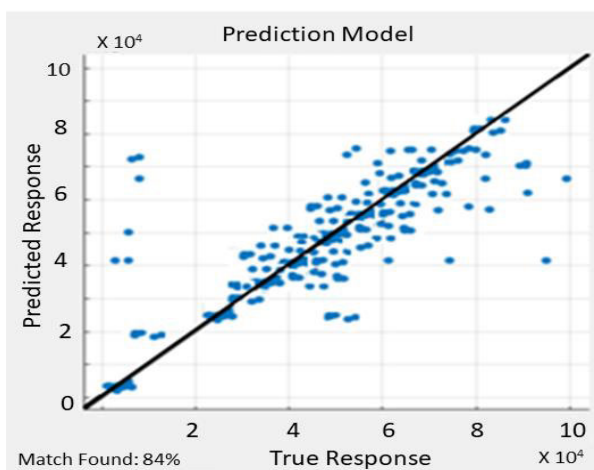


FIGURE 8. RMSE dispersion diagram.

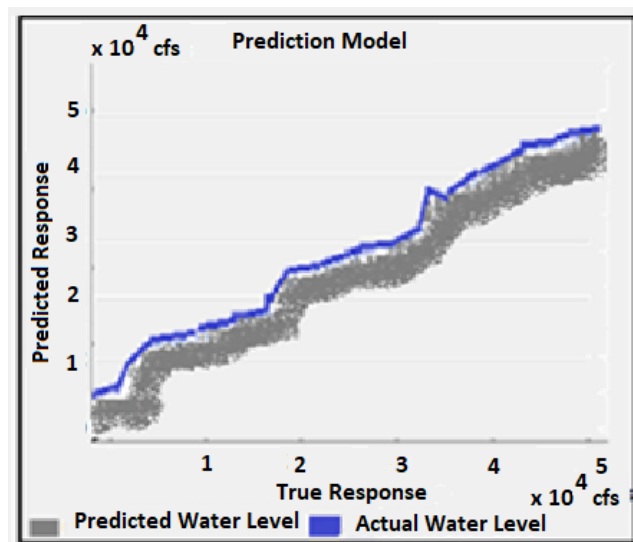


FIGURE 9. FFNN model performance in time (Per day).

peak error percentage on predicted water levels using validation data is less than 10 %. It proves that FFNN model has

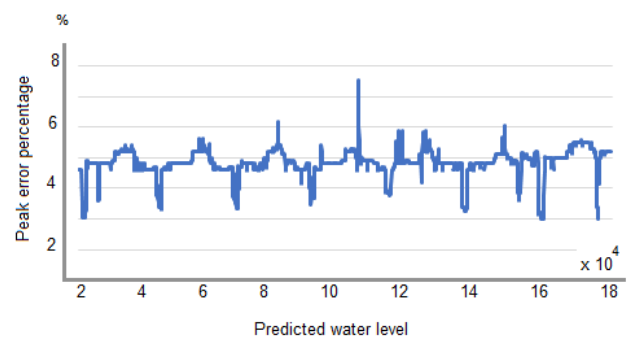


FIGURE 10. Peak error recorded on validation dataset.

successfully transformed input onto output and computed the result with 84 % accuracy. The results that are obtained from these models are compared in terms of accuracy, efficiency and time spent during these trainings.

VI. CONCLUSION

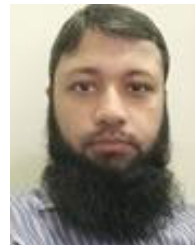
In this research, a flood forecasting model (FFM) has been presented that works in two modules. In first module, eighteen local stations have been monitored for training and transmitting local data models to central server. Central server trains global model that has been capable enough to determine the local station where flood is expected within net five days by analyzing multiple parameters extracted from local models. In second module of FFM, feed forward neural network model is trained at the local station where flooding has been predicted to determine the expected raise in water level during flood. Hydraulic and metrological data at eighteen local stations have been locally processes to preserve data privacy, guarantee data security and ensure data availability. The proposed FFM also issues flood alert to flood mitigation department for taking necessary actions towards disaster prevention and response. The proposed system has also been evaluated for prediction of historic floods encountered in the selected region from 2010 to 2015. The proposed FFS predicted the historical floods with 84% accuracy. Currently FFM has been trained on regional data of the selected zone

but in future, it can be expanded to predict floods in other regions of the world using their datasets.

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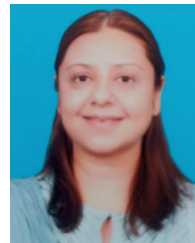
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